Using Statistical Methods from Econophysics to Analyse Bubble Motion in a Two-Dimensional Foam

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ABSTRACT

ACKNOWLEDGEMENTS

I want to thank my supervisor, Professor Stefan Hutzler, for his guidance and support throughout this project. The weekly meetings made a huge difference in the progress made in this capstone even if it took me a while to programme things on the fly. Thank you Stefan.

I would also like to thank Professor Simon Cox for providing the foam data that made this project possible, and a thank you to Professor Peter Richmond for this insight into the econophysics side of the project.

I am grateful to Professor Cox and Professor Richmond for the insightful meetings throughout the year, which helped monitor progress and discuss future aims of this work.

A special thanks to my family and friends for their patience and support throughout this college year.

Finally I would like to address the use of Gen AI in this project. I used a combination of ChatGPT and Claude AI not for any conclusion on data, but to allow me to use intuition and creativity while programming without the drawback of lack of niche Python syntax. I used AI as a crutch to accelerate progress rather than replace intuition and critical thinking which I believe is the correct way to use in a project like this.

Contents

1 Introduction

Foams are among the most familiar yet physically complex systems encountered in everyday life. From the froth on your morning coffee to the structure of biological tissues, foams occupy a unique place in the study of soft matter. At their core, a foam is a dispersion of bubbles seperated by a liquid medium, and the collective behaviour of these bubbles; how they coarsen over time and how the foam restructures because of this, is they core of what is being investigated in this report.

Coarsening is a fundamental property of these foams. It is the process by which bubbles exchange gas across their shared interfaces causing larger bubbles to get larger and smaller bubbles get smaller. Over time, the average bubble area increases and the number of bubbles decrease. During this coarsening, bubbles gain new neighbours as the smaller ones disappear causing topological rearrangements in the foam. Understanding the statistical character of the individual bubble trajectories, and how random they are, within this evolving foam structure is the main motivation of this project.

In this project, we use data from a 2D foam simulated by Prof. Simon Cox. In this data we are given the positions of the bubble centres at each timestep, along with individual properties such as area and coordination number. Rather than characterising the system through bulk statistical properties, we apply tools borrowed from econophysics to investigate the dynamics of this foam. The central analogy is between the bubble's displacements through the foam and the price movements in a financial market. In both cases, one is interested in the nature of the randomness involved. Whether it is completely random with simple Brownian diffusion, and whether there is a need to characterise a different type of randomness.

Brownian motion is used in this project as a baseline of randomness for comparison. It is well understood in it's statistical properties, Gaussian displacement distributions, mean squared displacement that scales linearly in time, and vanishing temporal autocorrelation.This means we can use it as a sanity check for all of our methods, if we apply our methods to it and get the expected results, we can be more confident in our results when we apply the same methods to the bubble trajectories. Deviations from this baseline are of particular interest as they suggest the presence of underlying statistical properties of the foam that influence individual bubble motion.

The report is structured as follows: Section 2 reviews the relevant foam physics, the statistical mechanics of Brownian motion, and the tools from econophysics we will be using; log-returns, volatility, temporal autocorrelation, etc. Section 3 describes the use of pandas to organise the data in a data-frame, extract information, and how we apply computational methods to this information. In section 4 we present our results and discuss their implications for understanding bubble motion in coarsening foams. Finally, in section 5 we summarise our findings and suggest directions for future research.

2 Background Theory

2.1 Foam Physics

coordination number

2.2 Brownian Motion

Brownian motion refers to the random movement of particles suspended in a fluid due to the cumulative effect of many independent collisions with the surrounding particles or molecules. This phenomenon was first observed by the botanist Robert Brown in 1827, who noticed pollen grains moving erratically in water.

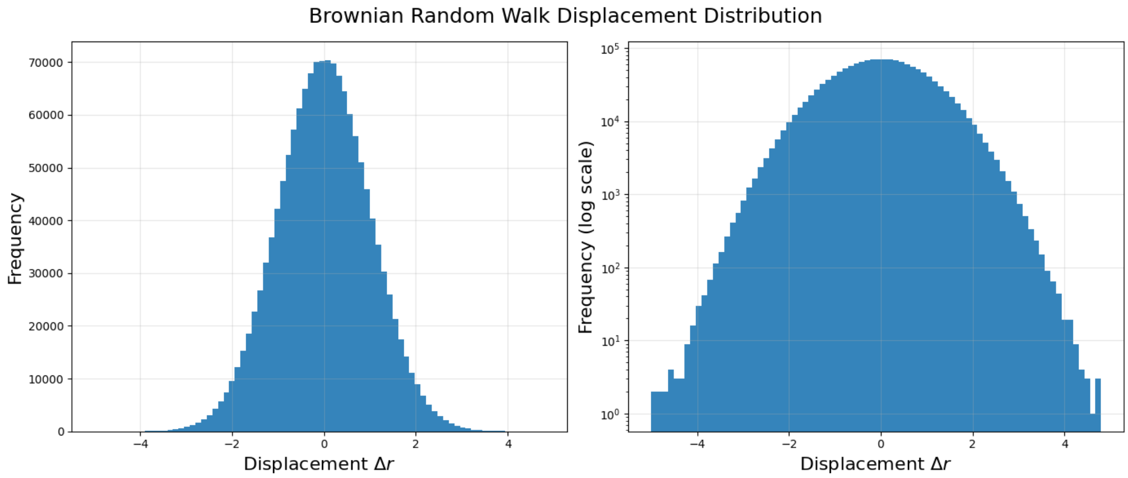


Figure 2.1: Averaging over 1000 Brownian random walks. As expected we get a Gaussian curve for the displacement distribution (left), and worth noting the inverted parabola on the log scale (right)

As mentioned in the introduction, since the statistical properties of Brownian motion are well known, we can use it as a sanity check to ensure our computational method is working accordingly. **Finish later**

2.3 Econophysics

In this section, I am going to first explain the application of these methods when analysing financial markets. Then I will explain how they intuitively can be applied to the bubble displacements to help us with our statistical analysis.

2.3.1 Log Returns

Suppose $s(t)$ is the price of an asset at time t. One can measure the price fluctuations by simply looking at the change in price over a time δt. [2]

$$\delta s(t, \delta t) = s(t + \delta t) - s(t) \tag{2.1}$$

However, the main problem with looking at the price change in an asset is that it is indirectly affected by external economic events; recession, inflation etc. It is more useful to look at the **log returns**.

$$r(t, \delta t) = \ln\left(\frac{s(t + \delta t)}{s(t)}\right) = \ln(s(t + \delta t)) - \ln(s(t)) \tag{2.2}$$

The log returns are now seen to be additive.

$$r(t, \delta t_1 + \delta t_2) = r(t + \delta t_1, \delta t_2) + r(t, \delta t_1) \tag{2.3}$$

The incorporation of an inflation factor, f(t), we get

$$\ln\left(\frac{f(t + \delta t)s(t + \delta t)}{f(t)s(t)}\right) = r(t, \delta t) + \ln\left(\frac{f(t + \delta t)}{f(t)}\right) \tag{2.4}$$

If we look at the second term, we see that if $f(t + \delta t) = f(t)$ then that term equals zero. The use of log returns takes account of inflation that remains constant over time. [2]

How this might apply to our bubbles is that we can use the log returns accountability of inflation to account for the overall growth of the bubbles in the foam over time as it coarsens. With this overall increase in bubble diameter the magnitude of the displacements increase aswell. For this reason, it is worth taking the log returns of the bubble displacements now instead so we can analyse the relative localised bubble dynamics rather than being affected by overall growth on top of that.

2.3.2 Volatility

In a financial market, volatility is a measure of the magnitude of price fluctuations. High volatility markets, such as cryptocurrencies, are characterised by large price swings, while low volatility markets, such as government bonds, tend to have more stable prices.

We can define the volatility as the absolute value of the log returns [2]:

$$v(t, \delta t) = |r(t, \delta t)| \tag{2.5}$$

This can directly be applied to our bubble displacements, intuitively applied as a measure of how strong the fluctuations in the magnitude of the bubble displacements are. We can use this to investigate whether there are periods of high volatility in the bubble motion, and whether these periods are correlated with other properties of the foam such as the coordination number or area of the bubbles.

2.3.3 Temporal Autocorrelation

A temporal correlation is a measure of how a variable x at time t is influencing its value at a time $t + \tau$. I always like the example of rolling across the room in a rolling chair. If I push off one wall while sitting in the chair, my position at that time is highly correlated with my position a second later, however that evening where I am writing my capstone thesis has nothing to do with whether I was rolling in a chair or not, in other words my position then is completely uncorrelated to my position when I rolled the chair.

This correlation measure is given by evaluating the *autocorrelation function*, $G(\tau)$, defined as [2]:

$$G(\tau) = \langle x(t)x(t+\tau)\rangle = \lim_{T\to\infty} \frac{1}{T}\int_0^T x(t)x(t+\tau)dt \tag{2.6}$$

This can be directly applied to our bubble sample. How long to the bubble displacements stay correlated for? Is there any correlation at all after a topologicalrestructure event?

2.4 Statistical Tools

2.4.1 Mean Squared Displacement

Mean Squared Displacement (MSD) is a measure of the average squared distance that a particle travels over a given time interval. It is commonly used in physics and chemistry to study the motion of particles in a fluid or gas. The MSD is defined as:

$$\mathrm{MSD}(\delta t) = \langle (\delta r(t,\delta t))^2\rangle = \frac{1}{N}\sum_{i=1}^{N}(\delta r_i(t,\delta t))^2 \tag{2.7}$$

where $\delta r_i(t, \delta t)$ is the displacement of the i-th particle at time t over a time interval δt, and N is the total number of particles. The MSD can provide insight into the underlying dynamics of the system, such as whether the motion is diffusive, subdiffusive, or superdiffusive. For example, in a purely diffusive system, the MSD scales linearly with time (MSD $\propto \delta t$), while in a subdiffusive system, the MSD scales slower than linearly (MSD $\propto \delta t^\alpha$ with $\alpha < 1$), and in a superdiffusive system, the MSD scales faster than linearly ($MSD \propto \delta t^\alpha$ with $\alpha > 1$). Rewrite Definition: tabbed through copilot suggestion

For Brownian motion, as described by Einstein in his 1905 paper , the MSD scaled linearly with time following the diffusion law [1]:

$$\langle r^2(\tau)\rangle = 2dD\tau \tag{2.8}$$

Where d is the dimension, D is the diffusion coefficient, and τ is the elapsed time. It is in our interest to see if the bubble MSD follows this diffusion laws alongside Brownian Motion or if there it follows another trend.

2.4.2 Displacement Distributions

The displacement of a bubble over a time δt is given by:

$$\delta r(t, \delta t) = r(t + \delta t) - r(t) \tag{2.9}$$

Analysing these displacements in different ways will give us insight into the underlying statistical properties of the bubble motion. In this case, we can make a displacement distribution and vary δt to see how the distribution changes looking at different timelags, looking at every timestep, every 10 timesteps, every 100 timesteps etc.

2.4.3 Complementary Cumulative Distribution Function

When investigating the log returns of a financial market, a key underlying statistical property of the probability density is that if we vary Δt the distribution scales according to a Levy distribution.

Figure 2.2a is the probablility distribution functions for the log returns over different Δt, all overlayed on one plot with the axes shifted for visual purposes. The complementary cumulative distribution function (CCDF), see figure 2.2b is obtained by integrating the probability distribution functions (PDF) for $|x| > x$ giving the probability the log returns are greater than a certain value for each Δt PDF curve.

A property of scaling according to a Levy distribution is that if you take a cut at a certain probability on the CCDF curve, is that it is independent of where the cut is taken as seen in figure 2.2c where the cut, normalised to the $\Delta t = 1$ cut, increases linearly with Δt.

Finally, the slope of this linear plot can be used to scale the x axis of the original plot (2.2a) by plotting over the x axis, $x' = x/\Delta t^{\beta}$, resulting in the collapse of all PDFs onto one *master curve* (Figure 2.2d). A Levy curve can be fit to the peak of this plot using a Levy fit parameter of $\alpha = 1/\beta$

It would be interesting to apply to the bubble displacements and log returns to see if they also scale according to a Levy distribution, if so, what would the scaling exponent be? If not, it is worth investigating what he underlying scaling properties are. One consideration before we perform this analysis is that a key difference between these financial markets and the foam is that as the foam coarsens the number of bubbles decrease over time, whereas there is constantly an abundance of data for the financial log returns.

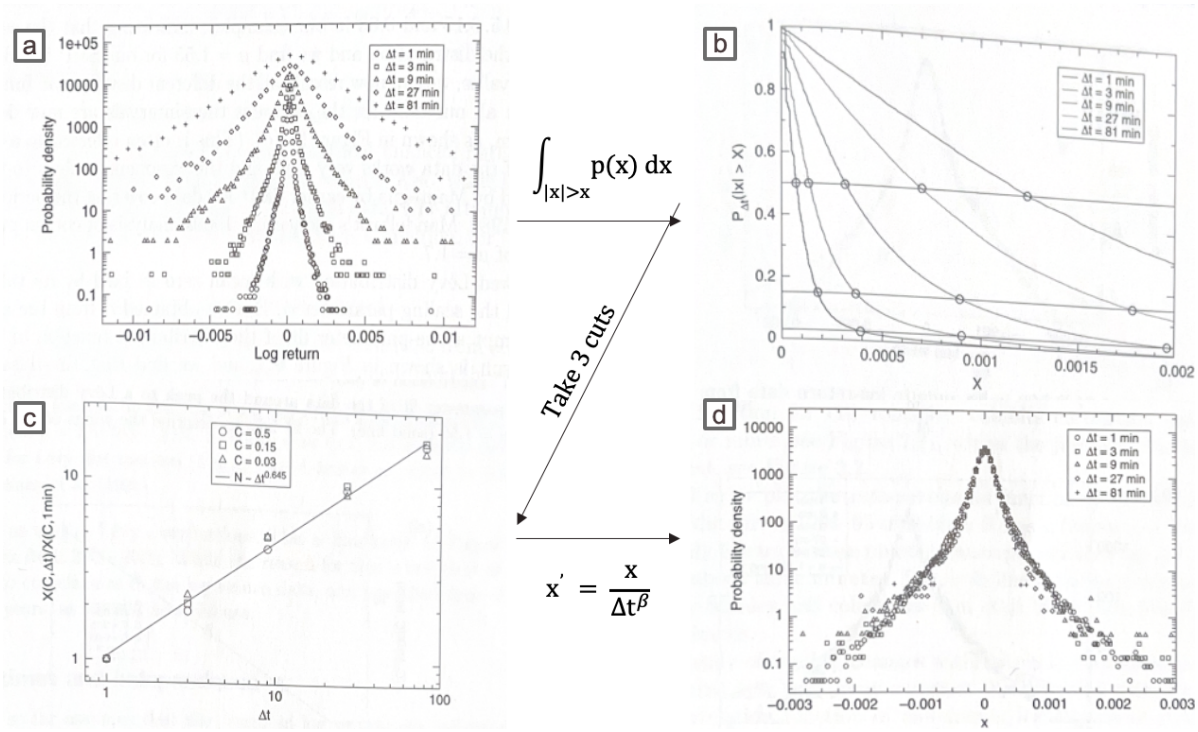


Figure 2.2: Series of figures which describes how complementary cumulative distribution functions can be use to collapse data onto a master plot. [2]

3 Computational Methods

4 Results and Discussion

4.1 Bubble Paths

Figure 4.1 shows the path of the centres of the bubbles during the simulation. The mean position of the bubbles is shown in the centre of the plot to investigate for an overall drift in the bubbles. Mainly, this is to prove that there is no bias in displacements in the x and y direction, so when we look at the distribution of displacements, we can combine the x and y displacements for double the data.

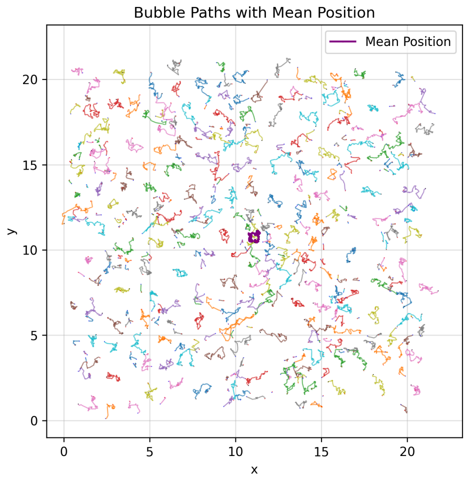


Figure 4.1: Paths of the centres of the bubbles.

Figure 4.2a shows a Brownian random walk over 726 steps, the same number of time steps in the foam simulation. Figure 4.2b shows the path of one of the bubbles which survives the full 726 timesteps. Comparing the two side by side, it is clear that the Brownian random walk is completely random, erratic, and uncorrelated. Whereas the bubble path seems slightly more structured, with some steps clustered together, when the foam is relaxed, and some large jumps then the foam is restructuring. This is a stark difference between the two, however it is worth comparing with the Brownian walk as a baseline to see how random, or what particular type of randomness, the bubble paths follow.

4.1.1 Average Area

As a sanity check, the average area over time is shown to be linear in figure figure. This was proven in two ways. The first was using the raw diameter data and averaging that the area obtained from that over all the bubbles in that timestep, The second method used an approximation, $\langle A \rangle = A_{box}(1 - \phi)/N(t)$, where ϕ is the liquid fraction and $N(t)$ is the number of bubbles at time t. Figure figure proves these two methods line up exactly.

The key takeaway from this is that since the area is increasing linearly with time then we are losing bubbles while this is happening.

4.1.2 Mean Squared Displacement

The mean squared displacement (MSD) was averaged over 400 Brownian random walks, and as expected scaled linearly with time. In figure 4.4, this Brownian MSD line is superimposed onto the plot of the bubbles MSD over time. There are some stark differences. Initially, since I was looking for this linear relation in the bubbles I tried to convince myself there was some

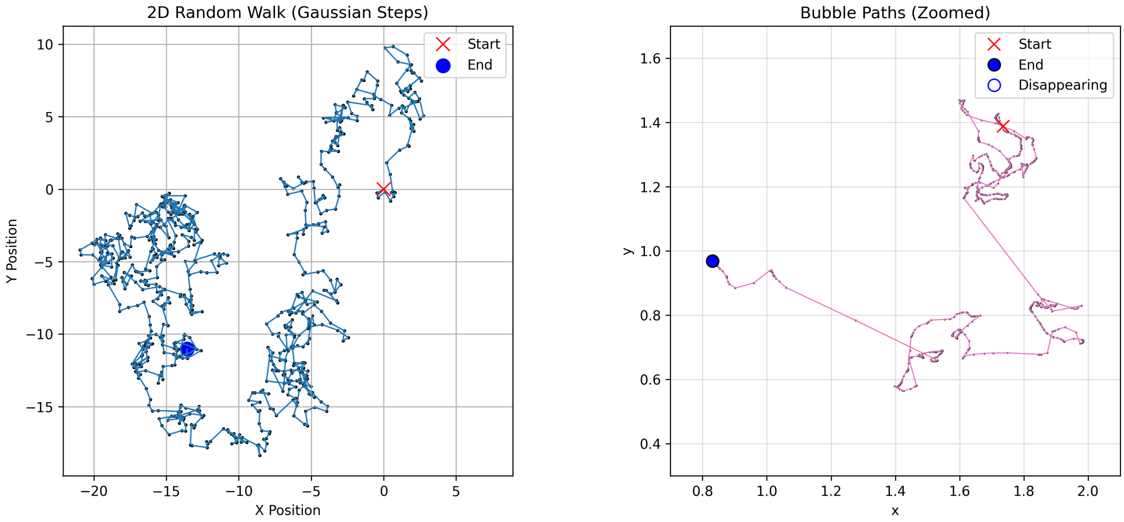


Figure 4.2: Comparison of a Brownian random walk (left) and a tracked bubble path (right) over the same number of timesteps.

sort of linearity in the plot. While there certainly is an upward trend, it is definitely not linear, at least not with the amount of data we have.

One clear distinction between the two simulations is that the number of bubbles decrease over time. This can be seen in the figure 4.4 with the red dashed lines marking the milestones in the bubble counts. The less data you have the less linear the plot would be regardless, this can be seen once half the bubbles have disappeared around the 260th timestep. However even when we have the most amount of bubbles, 400-300 bubbles between t=0 and t=80, this section of the plot almost looks like it follows a power law.

The conclusion from this investigation is that it is hard to make any good conclusions from this analysis. For that reason, we move on to other method to analyse the statistics of this system.

4.1.3 Displacement Distributions

Brownian Baseline add later The displacemnt distribution of the bubble displacements looking at every timestep ($\Delta t = 1$) has a sharp peak around 0 and tapers down each side, as seen in figure 4.5 on a log-lin scale. Comparing with the Brownian displacement curve in figure 2.1, there is a stark difference between the two distributions; an inverted parabola for the Brownian motion, and a sharp peak that tapers off each side for the bubble displacements.

As Δt increases the bubble displacement distributions widen, eventually approaching a Gaussian shaped inverted parabola. This shows the uncorrelation comparing displacement at a certain timestep versus displacement a hundred or so timesteps later. There is an element of

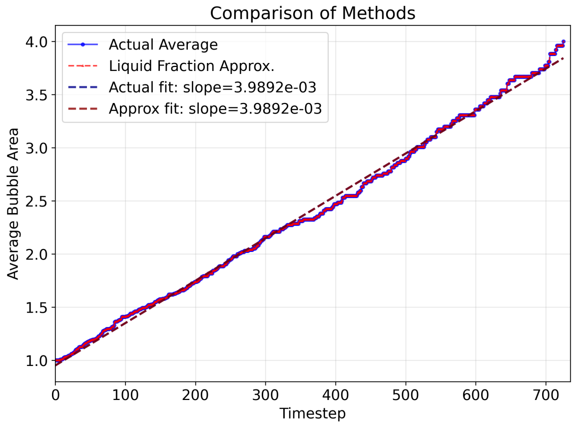


Figure 4.3: Average bubble area increases linearly over time.

randomness which brings the Gaussian distribution into play.

Investigating the log return distributions led to a different conclusion than the displacement distributions. Unlike the displacement distributions, the log return distribution does not widen into a Gaussian curve. As the distribution don't evolve into a Gaussian, section 4.1.5 will investigate if there is an element of correlation in the log returns.

4.1.4 Complementary Cumulative Distribution Functions Collapse

Brownian Baseline add later

Displacement distributions from the section 4.1.3 were used for an initial attempt at a Complementary Cumulative Distribution Function (CCDF) collapse.

A key difference between the Brownian scaling and the scaling of financial markets is that the CCDF cuts are dependent on where the cut is taken, there is not a linear relation between the cut (normalised to the first cut) versus Δt. For arguments sake, the middle slope is taken, $\beta = 0.63$ giving a relatively good collapse in figure 4.8. The centre of the distribution collapses well, however, the tails clearly deviate from this scaling a lot.

To investigate this scaling further, more Δt distributions were added along with more CCDF cuts in an attempt to fit a function to these scaling properties. *See appendix: figure 6.1*. Figure 4.9 shows the resulting scaling of each cut with Δt.

By process of trial and error to find a function which scales this way, equation 4.1 seemed the best fit.

$$f(\Delta t) = \frac{c_1 \Delta t}{1 - c_2 \Delta t} \tag{4.1}$$

Equation 4.1 uses two free parameters, c_1 and c_2, and the fit uses a least square method to obtain the corresponding c_1 and c_2 for the best fit. Each free parameter value is stored alongside the

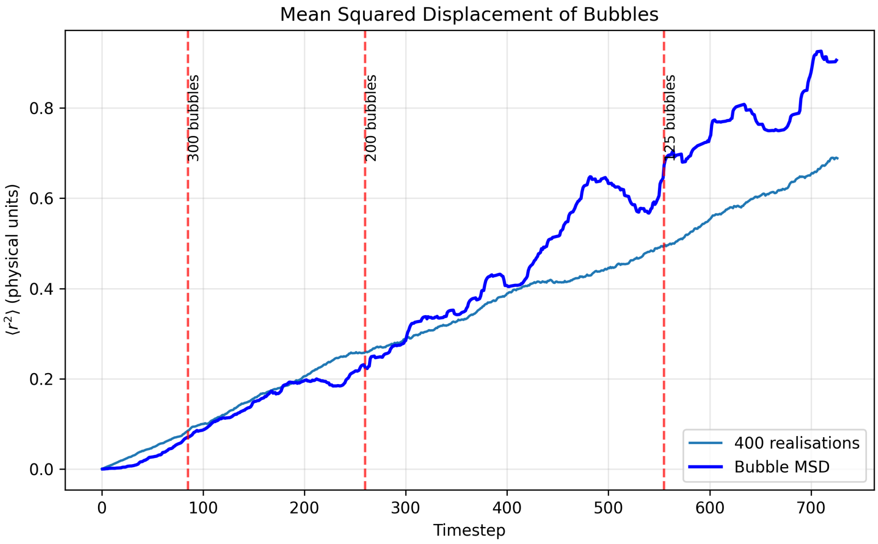


Figure 4.4: Mean squared displacement for the bubble data with the MSD for 400 Brownian walks overlayed to compare linearity.

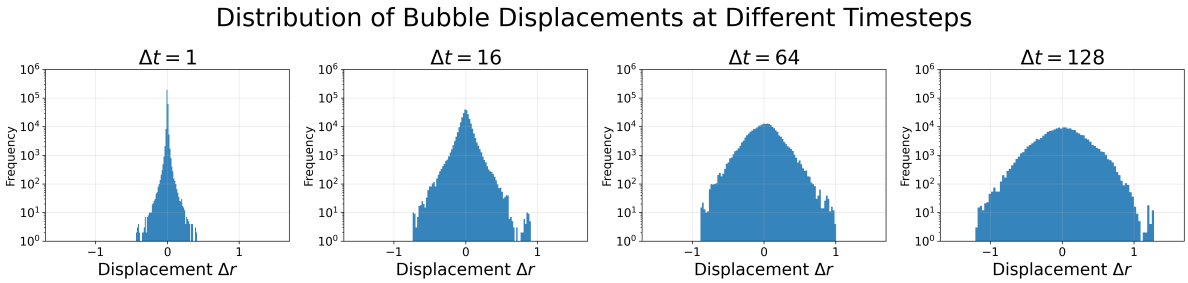


Figure 4.5: Displacement distributions looking at different time lags. Distibutions widen towards a Gaussian distribution at larger time lags.

cut C that these values are associate with. Plotting c_1 against C and $1/c_2$ against C gives a linear relation for each as shown in figure 4.10.
given more time I am hoping to somehow use this slope and $C = 0$ value to scale the displacement distribution PDFs onto one masterplot.
Peak scaling method, appendix maybe

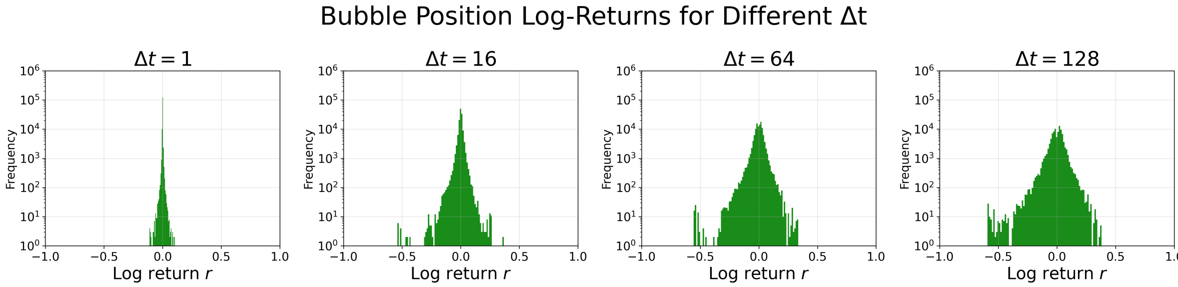


Figure 4.6: Log return distributions over increasing Δt. Distribution does not widen into a Gaussian unlike the displacement distributions. Log-linear scale.

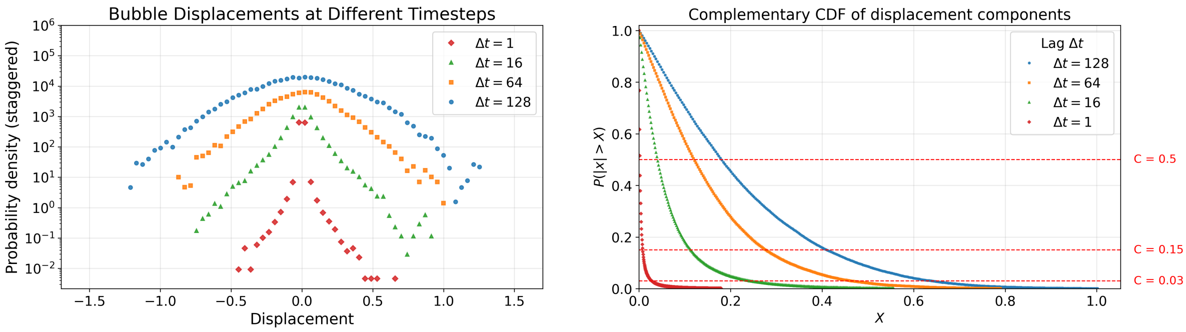


Figure 4.7: Displacement distributions overlayed on the one plot, axes shifted for visual purposes (left). These PDFs were then integrated over $|x| > x$ to get the CCDF (right).

4.1.5 Autocorrelation Functions

5 Conclusion

References

[1] A. Einstein. *Über die von der molekularkinetischen Theorie der Wärme geforderte Bewegung von in ruhenden Flüssigkeiten suspendierten Teilchen*. Annalen der Physik (in German), 1905.

[2] P. Richmond, J. Mimkes, and S. Hutzler. *Econophysics and Physical Economics*. Oxford, UK: Oxford University Press, 2013.

6 Appendix

6.1 CCDF Scaling

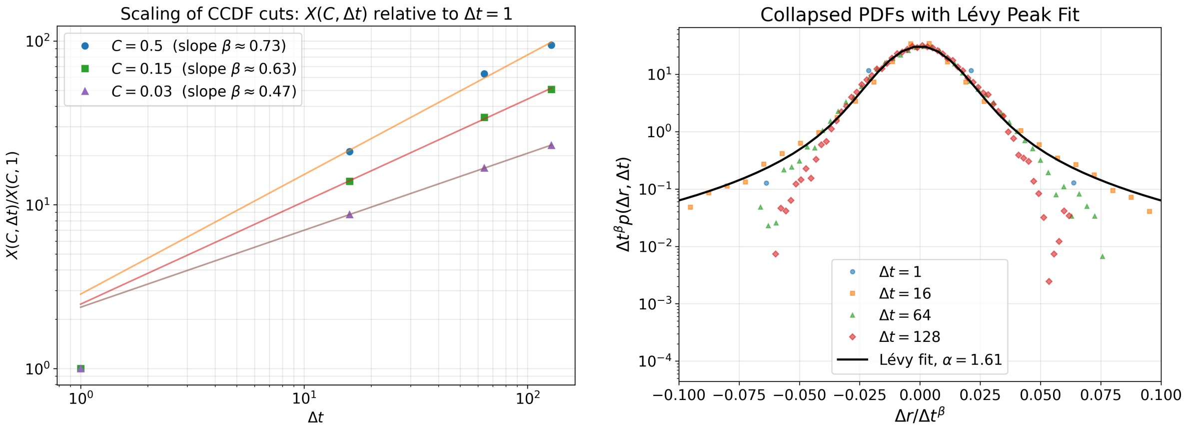
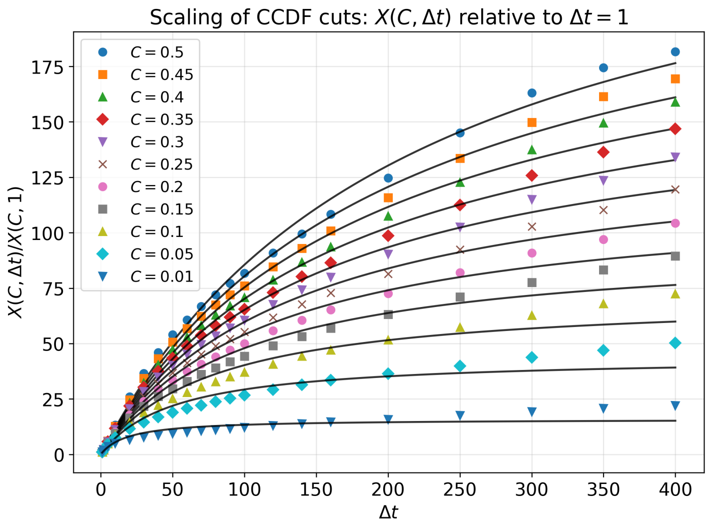


Figure 4.8: Bubble displacements do not scale according to Levy distribution.



C	c_1	c_2
0.50	1.240	0.00455
0.45	1.200	0.00497
0.40	1.160	0.00538
0.35	1.110	0.00589
0.30	1.080	0.00653
0.25	1.040	0.00737
0.20	0.998	0.00848
0.15	0.962	0.01010
0.10	0.920	0.01290
0.05	0.870	0.01970
0.01	0.829	0.05210

Figure 4.9: Bubble displacements do not scale according to Levy distribution (left). Fitted parameters for each CCDF cut (right).

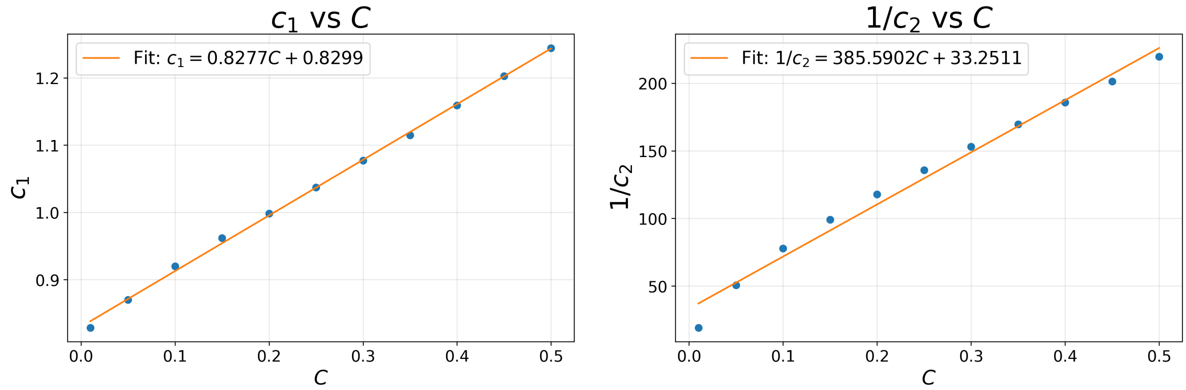


Figure 4.10: Linear relation for scaling of c_1 and $1/c_2$ with respect to C

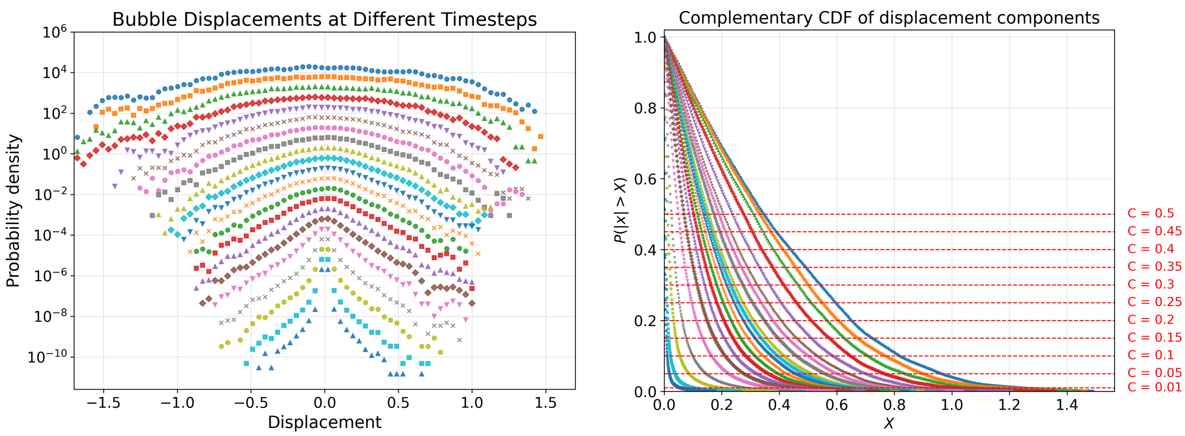


Figure 6.1: Bubble displacements do not scale according to Levy distribution.